TokenTriage: Harm-Aware Token Routing for Clinical Language Model Inference

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June 2026

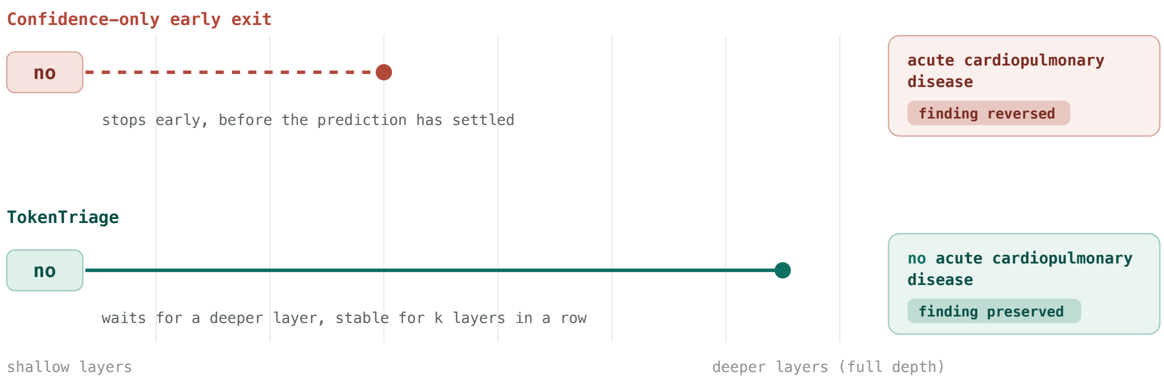

Figure 1. Overview of TokenTriage. The clinically critical word *no* is decoded two ways by the same model. A confidence-only early exit stops at a shallow layer, before the prediction has settled, and drops the negation, so the report reads *acute cardiopulmonary disease* and the finding is reversed. TokenTriage holds the same word until a deeper layer at which the top prediction has been stable for k consecutive layers, which preserves *no* and the original finding. Ordinary words still take the cheap early exit, so most of the compute saving is kept.

Abstract

Adaptive-inference methods such as early exit reduce language-model compute by stopping computation at an intermediate layer when the prediction for a token already looks confident. We argue that for clinical text, an average compute-quality trade-off is the wrong thing to optimize, because clinical language is semantically asymmetric. An early error on *no*, *left*, *severe*, a measurement, or a disease term can invert the meaning of a report, while an early error on a function word is harmless. We formalize a harm-stratified view of adaptive inference and measure preservation of the full model at the token level, conditioned on clinical semantic risk. Across four open models (Qwen2.5-0.5B/1.5B/3B and SmolLM2-1.7B; two architecture families and a 6x parameter range) and two public radiology datasets, the strongest confidence-only early exit, at a compute saving of 37.9%, disagrees with the full model on 71.7% of clinically critical tokens on 256 IU X-Ray reports. We introduce TokenTriage, a routing rule that leaves ordinary tokens to a confidence exit but requires critical tokens to reach a minimum depth and k consecutive stable predictions. It reduces critical-token disagreement to 24.9% (95% CI $[0.226, 0.276]$) at comparable savings. We validate the effect against ground truth in two ways. Measured against the exact words of the radiologist, confidence-only exit introduces 16 to 22 points of excess critical-token error over the full model, while TokenTriage introduces 0 to 4. A CheXbert counterfactual shows that TokenTriage roughly halves both the reports whose clinical findings change (70% down to 46%) and the total findings flipped (400 down to 224). Finally, we report a negative result that places the contribution honestly: untrained free-running early-exit generation collapses for all policies, so TokenTriage belongs in the preservation and verification regime, most directly as harm-aware acceptance for self-speculative decoding, rather than as a standalone generation speedup. Code, traces, and analyses are released openly.

1 Introduction

A transformer language model spends the same computation on every token, regardless of how much that token matters. Adaptive-computation methods use the observation that many tokens are easy, in the sense that their final prediction is already settled at an intermediate layer, and stop early to save latency and cost [26, 28, 19, 23, 8]. The usual exit criteria are local measures of model confidence: the entropy of the intermediate distribution, the top-1 minus top-2 margin, or the response of a trained early-exit classifier.

Confidence is a property of the model, not of the cost of being wrong. In clinical documentation the two come apart. Consider the phrase "*no* acute pulmonary edema." The token *no* may be low or high entropy at a given layer, but its semantic leverage is large: an early commitment to the wrong token here reverses the finding. The same holds for laterality (*left* or *right*), severity (*mild* or *severe*), temporality (*new* or *chronic*), numeric measurements, and named findings. A function word like *the* carries no such leverage. An efficiency method that allocates compute by confidence alone is therefore blind to the dimension that matters clinically.

This paper asks a narrow question. When adaptive inference saves compute on clinical text, is the saving drawn disproportionately from the tokens whose errors carry clinical risk? We answer it by measuring preservation of the full-model prediction at the token level, stratified by a semantic risk label, and, because preservation is not the same as correctness, also against the gold radiologist text and against a clinical entity labeler. We then propose a small intervention that reallocates compute by risk rather than by confidence alone.

Scope: where adaptive depth fits. We want to be clear about deployment relevance. Post-training quantization [6, 10, 18] and distillation are the dominant techniques in use today for fast LLM inference, and they reduce cost on a different axis (numerical precision and model size) than adaptive depth. Early exit is complementary to both, since a quantized model can still exit early, and its most widely used relative is self-speculative decoding, in which early-exit drafts are verified by the full model [8]. We do not claim that adaptive depth should replace quantization. The contribution is a harm-aware lens, and a routing or verification rule, for the adaptive-depth family when it is applied to clinical text, where the question of which drafts to trust without a full-model check is itself a semantic-risk decision.

Contributions.

1. A harm-stratified evaluation of adaptive inference (Section 3): an objective that conditions preservation of the full model on clinical semantic risk, which exposes a failure mode that average compute-quality curves hide.

2. TokenTriage (Section 4, Algorithm 1): a simple, training-free routing rule that combines a confidence exit for ordinary tokens with a depth and k-stability requirement for critical tokens, together with a pluggable risk-tagger interface.

3. Evidence across models, datasets, and ground truth (Section 6): confidence-only adaptive inference concentrates error on critical tokens across two model families and a 6x size range, on a second dataset, against the exact words of the radiologist, and, through a CheXbert counterfactual, at the level of clinical findings.

4. An honest decoding-regime analysis (Section 6.5): untrained free-running early-exit generation collapses regardless of policy, which places the contribution in the preservation and verification regime and connects it to self-speculative decoding.

2 Related Work

Efficient inference. The dominant production techniques reduce precision or size: post-training quantization to 8-bit [6] and to 3 or 4 bits [10, 18], knowledge distillation [13, 22], and one-shot pruning [9]. These act on a different axis from the depth axis studied here and compose with it, which is why we treat adaptive depth as complementary rather than competing.

Adaptive computation and early exit. Adaptive computation time learned to spend a variable number of steps per token [11], and PonderNet reframed this as a learned halting distribution [2]. For transformers [27], BranchyNet attached side classifiers so confident inputs could exit early [26], the depth-adaptive transformer learned per-token exit depths [7], and DeeBERT, FastBERT, PABEE, and the right-tool-for-the-job line brought input-adaptive exiting to encoders by confidence or patience [28, 19, 29, 24]. CALM extended adaptive computation to autoregressive generation with calibrated local confidence measures and a consistency guarantee [23], while mixture-of-depths routes a compute budget across positions [21]. LayerSkip trains a model with layer dropout and an early-exit loss so that early exit and self-speculative decoding work without auxiliary heads [8]. All of these select exits by confidence, calibration [12], or a trained acceptance criterion, and none condition on the semantic consequence of the token, which is our focus.

Speculative and self-speculative decoding. Speculative decoding speeds up generation by drafting with a cheap model and verifying with the target model [16, 5], and variants such as Medusa and EAGLE add lightweight draft heads [4, 17]. Self-speculative decoding uses the early layers of the model as the drafter [8]. This is the most realistic deployment path for adaptive depth, and it is a setting where a harm-aware acceptance rule, one that trusts cheap drafts for ordinary tokens but asks for full-model confirmation on critical ones, is natural.

Clinical report understanding. RadGraph extracts clinical entities and relations from radiology reports [15], and CheXbert [25] labels reports with the 14 standard CheXpert chest-radiograph findings [14]. Both are used to evaluate clinical text generation. We use CheXbert to convert token-level error into a clinical-finding-level measure of harm (Section 6.4). To our knowledge, prior efficiency work for clinical LLMs has not stratified adaptive-inference error by semantic risk.

3 Problem Formulation

Let a report be a token sequence $\boldsymbol{x} = (x_1, \ldots, x_n)$ processed by an L-layer transformer. For position i, let $\boldsymbol{h}_i^{(\ell)}$ be the hidden state after layer ℓ. The logit lens, an unembedding probe of intermediate states [3], projects a state to the vocabulary using the final norm of the model LN and unembedding W_U:

$$\boldsymbol{z}_i^{(\ell)} = W_U \, \mathrm{LN}\big(\boldsymbol{h}_i^{(\ell)}\big), \qquad \boldsymbol{p}_i^{(\ell)} = \mathrm{softmax}\big(\boldsymbol{z}_i^{(\ell)}\big). \tag{1}$$

Write the layer-ℓ prediction $\hat{y}_i^{(\ell)} = \arg\max_v \boldsymbol{p}_i^{(\ell)}(v)$, and the full-model prediction $\hat{y}_i^{(L)}$. An exit policy π assigns each position an exit layer $e_i = \pi(i) \in \{1, \ldots, L\}$ and emits $\hat{y}_i^{(e_i)}$. Its simulated compute saving is $1 - \frac{1}{nL}\sum_i e_i$.

Semantic risk. A risk tagger c, mapping each token to a value in $[0, 1]$, assigns each position a criticality $c_i = c(x_i)$, and the set of critical positions is $\mathcal{C} = \{i : c_i \geq \tau_c\}$ with $\tau_c = 0.8$. We treat c as a pluggable interface (Section 4.2).

Harm-stratified preservation. The quantity of interest is not average accuracy but where a policy deviates from the full model. We define critical-token disagreement

$$\mathrm{D}_{\mathcal{C}}(\pi) \;=\; \frac{1}{|\mathcal{C}|}\sum_{i\in\mathcal{C}} \mathbb{1}\!\left[\hat{y}_i^{(e_i)} \neq \hat{y}_i^{(L)}\right], \tag{2}$$

and harm-weighted disagreement $\mathrm{HWD}(\pi) = \left(\sum_i c_i \mathbb{1}[\hat{y}_i^{(e_i)} \neq \hat{y}_i^{(L)}]\right) / \sum_i c_i$. A good clinical policy minimizes $\mathrm{D}_{\mathcal{C}}$ at a given compute saving, even at the cost of higher disagreement on ordinary tokens.

4 Method

4.1 Layerwise tracing

We run each report teacher-forced once and cache, for every (i, ℓ), the normalized entropy, the top-1 minus top-2 margin, the target negative log-likelihood, the top token $\hat{y}_i^{(\ell)}$, and the agreement indicator $\mathbb{1}[\hat{y}_i^{(\ell)} = \hat{y}_i^{(L)}]$:

$$H_i^{(\ell)} = \frac{-1}{\log|V|}\sum_v \boldsymbol{p}_i^{(\ell)}(v)\log \boldsymbol{p}_i^{(\ell)}(v), \qquad M_i^{(\ell)} = \boldsymbol{p}_i^{(\ell)}{}_{(1)} - \boldsymbol{p}_i^{(\ell)}{}_{(2)}. \tag{3}$$

Caching separates tracing from policy evaluation, so every policy below is scored on identical traces, which makes the comparisons exact and cheap.

4.2 Risk assignment (pluggable)

The default tagger is a clinical lexicon that flags negation, laterality, severity, temporality, medication, numeric measurement, measurement unit, and key findings (for example `no`, `left`, `severe`, `2.4`, `cm`, `effusion`). The tagger is an interface: any function that maps a token to a value in $[0, 1]$, including a RadGraph- or CheXbert-derived tagger, can be registered without modifying the routing or evaluation code.

4.3 Policies

We evaluate four reference policies and TokenTriage. The entropy exit stops at the first layer with $H_i^{(\ell)} \leq \tau$, the margin exit at the first with $M_i^{(\ell)} \geq \gamma$, fixed-depth at $\lceil \rho L \rceil$, and a non-causal stability oracle at the first layer after which $\hat{y}_i^{(\ell)} = \hat{y}_i^{(L)}$ for all deeper layers, which is an upper bound on achievable preservation.

4.4 TokenTriage

TokenTriage (Algorithm 1) applies the entropy exit to ordinary tokens but holds critical tokens to two extra requirements: a minimum depth $\lceil \rho L \rceil$, and k-stability, meaning the top prediction must be unchanged for k consecutive layers. The second requirement is evidence that the early decision is not an artifact of an unsettled intermediate representation. Unless stated otherwise we use $\tau = 0.10$, $\rho = 0.85$, and $k = 4$.

Algorithm 1 TokenTriage exit decision for position i

Require: per-layer top tokens $\hat{y}_i^{(1:L)}$, entropies $H_i^{(1:L)}$, criticality c_i; threshold τ, min. fraction ρ, patience k

$b \leftarrow \min\{\ell : H_i^{(\ell)} \leq \tau\}$ (set $b \leftarrow L$ if none) ▷ confidence exit
if $c_i < \tau_c$ **then**
    return b ▷ ordinary token: cheap exit
end if
$\ell_{\min} \leftarrow \lceil \rho L \rceil$
for $\ell = \max(b, \ell_{\min}, k)$ **to** L **do**
    if $\hat{y}_i^{(\ell-k+1)} = \cdots = \hat{y}_i^{(\ell)}$ **then**
        return ℓ ▷ k-stable
    end if
end for
return L

4.5 Correctness and entity-level metrics

Beyond preservation we report two ground-truth metrics. Let g_i be the gold (radiologist) token. The excess critical-token error is the extra error a policy makes over the full model, $\Delta(\pi) = \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \left(\mathbb{1}[\hat{y}_i^{(e_i)} \neq g_i] - \mathbb{1}[\hat{y}_i^{(L)} \neq g_i] \right)$, which removes the high intrinsic floor of exact next-token prediction. The entity-level harm metric (Section 6.4) replaces each critical position in the gold report with the prediction of the policy and labels the gold and perturbed reports with CheXbert, counting changes to the 14-dimensional finding vector.

5 Experimental Setup

Data and models. The primary experiment uses the public `ChayanM/IUXray-Data-Train-Test` test split (256 reports, 13,697 next-token positions, 1,163 critical tokens, L=28) with Qwen2.5-1.5B-Instruct [20]. We test generalization on Qwen2.5-0.5B/1.5B/3B ($L = 24/28/36$) and SmolLM2-1.7B [1] ($L = 24$) at 64 reports, and on a second dataset, `Santhosh1705kumar/radiology-reports-chest` (ROCO chest captions, 64 reports). All data and models are public.

Protocol. Hidden states are computed in `float16` on Apple MPS. Confidence intervals are report-level bootstraps (200 resamples), resampling reports to respect within-report correlation. The compute saving is the simulated layer fraction $1 - \bar{e}/L$. We sweep $\tau \in \{0.10, \ldots, 0.55\}$ and $\gamma \in \{0.05, \ldots, 0.50\}$ and report the strongest confidence baseline by $\mathrm{D}_{\mathcal{C}}$.

6 Results

6.1 Confidence-only exit concentrates error on critical tokens

The strongest confidence baseline (entropy, τ=0.10) saves 37.9% of simulated compute yet disagrees with the full model on 71.7% of critical tokens (Table 1). TokenTriage at the matched threshold reduces $\mathrm{D}_{\mathcal{C}}$ to 24.9%, a 46.9 point absolute reduction, while retaining 33.1% savings, and the bootstrap intervals do not overlap. A higher-savings operating point (τ=0.20) reaches 51.7% savings at 27.0%, still well below the baseline.

Method	Setting	Compute saved	$D_{\mathcal{C}}$	95% CI	HWD
Entropy	τ=0.10	0.379	0.717	[0.686,0.747]	0.560
Margin	γ=0.50	0.571	0.865	[0.848,0.884]	0.779
TokenTriage	τ=0.10, ρ=0.85, k=4	0.331	**0.249**	[0.226,0.276]	0.390
TokenTriage	τ=0.20, ρ=0.85, k=4	0.517	0.270	[0.247,0.297]	0.575

Table 1. Main result on 256 IU X-Ray reports with Qwen2.5-1.5B. Confidence-only exits save compute but concentrate disagreement on critical tokens. TokenTriage more than halves it at comparable savings, and the result is stable versus the 64-report run (0.721 down to 0.255).

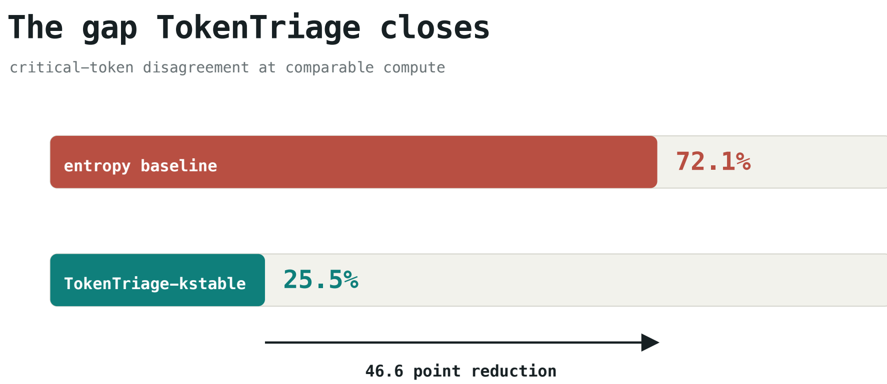

Figure 2. The same comparison on the 64-report sample. At comparable compute savings, the confidence-only entropy exit disagrees with the full model on the majority of critical tokens, while TokenTriage closes most of the gap. The model and the traces are identical; only the halting rule for high-risk tokens differs.

6.2 Generalization across families, scale, and dataset

6.3 Correctness against the words of the radiologist

Because the teacher-forced text is the gold report, exact next-token match is a ground-truth signal. Its floor is high, since free prose admits synonyms, so we report the excess $\Delta(\pi)$ rather than the raw rate. Table 3 and Figure 4 show that confidence-only exit adds 16 to 22 points of excess critical-token error against the words of the radiologist, while TokenTriage adds 0 to 4.

6.4 Entity-level correctness with CheXbert

Not every token error is clinically meaningful. We test the impact directly: we apply the critical-token errors of each policy to the gold report and label the gold and perturbed reports with CheXbert [25], then count changes to the finding vector. On the 256-report run, the critical-token errors of confidence-only exit change at least one finding in 69.9% of reports and flip 400 finding-labels overall. TokenTriage reduces this to 46.1% and 224, a 44% reduction (Table 4, Figure 5). TokenTriage does not eliminate clinical change, which is the honest residual, but it roughly halves it.

6.5 The decoding regime: an honest negative

The results above hold in the preservation regime, where the conditioning text is fixed (teacher forcing, scoring, grounded generation, and speculative-decoding verification). We tested whether harm-aware routing also helps during free, autoregressive generation by implementing real early-exit decoding, in which each emitted token is chosen at its exit layer and fed back, so that errors

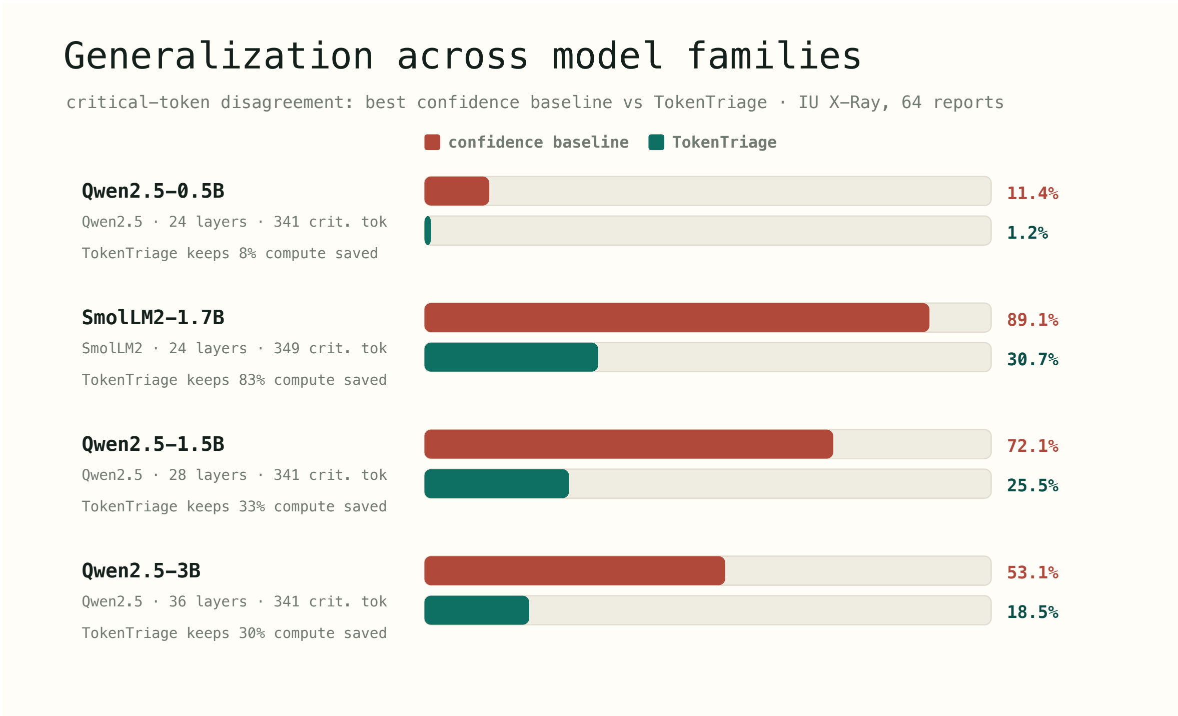

Figure 3. Critical-token disagreement across four open models (two families, 0.5B to 3B) on the same IU X-Ray reports. TokenTriage (teal) is well below the confidence baseline (clay) in every case. Smaller models show a smaller gap, because when the baseline is already safe there is little to fix.

propagate. We compared full, entropy, and TokenTriage decoders on 24 IU X-Ray continuations (Qwen2.5-1.5B, τ=0.10). The finding is negative. Untrained logit-lens early exit collapses for both policies (prefix agreement with the full model near 2%, critical-term fidelity near 0.10), and TokenTriage changed the emitted token in 0 of 7 critical interventions. When a model freely generates a critical token it is already confident, so the cheap-exit token is stable and harm-gating does not fire. This is consistent with why deployed systems train exit heads [23, 8] rather than read a raw logit lens, and it locates the value of TokenTriage where the target text is given, namely verification and self-speculative-decoding acceptance, rather than in naive untrained generation.

7 Discussion

Our results support a single claim: the compute-quality frontier is the wrong object for clinical adaptive inference, because it averages over a semantically asymmetric token distribution. A policy can look attractive in aggregate while concentrating both model-preservation error and ground-truth error on the tokens that determine the meaning of the report. TokenTriage shows that a small, training-free change to the halting rule, spending a few extra layers only on high-risk tokens, removes most of that concentrated error, and the effect is consistent across two model families, a sixfold range of model size, a second dataset, the words written by the radiologist, and the findings recovered by CheXbert.

We do not read this as an argument that adaptive depth should replace quantization. Quantization and distillation reduce cost on the axis of numerical precision and model size, adaptive depth reduces it on the axis of computation per token, and the two compose: a quantized model can still exit early. The most realistic deployment for adaptive depth is therefore self-speculative decoding, where early-exit drafts are accepted if they match the full model and otherwise corrected.

Model / setting	Family	$\lvert\mathcal{C}\rvert$	Baseline $D_{\mathcal{C}}$	TokenTriage	Saved
Qwen2.5-0.5B (64)	Qwen2.5	341	0.114	**0.012**	0.085
Qwen2.5-1.5B (64)	Qwen2.5	341	0.721	**0.255**	0.329
Qwen2.5-3B (64)	Qwen2.5	341	0.531	**0.185**	0.297
SmolLM2-1.7B (64)	SmolLM2	349	0.891	**0.307**	0.830
Qwen2.5-1.5B (256)	Qwen2.5	1163	0.717	**0.249**	0.331
ROCO / Qwen2.5-1.5B (64)	Qwen2.5	150	0.620	**0.153**	0.273

Table 2. Generalization (top), plus scale and a second dataset (bottom). The pattern is consistent, and the 256-report run confirms it is not a small-sample artifact.

Model	Full-model floor	Entropy excess Δ	TokenTriage excess Δ
Qwen2.5-0.5B	0.771	+0.021	**+0.000**
Qwen2.5-1.5B	0.757	+0.167	**+0.024**
Qwen2.5-3B	0.733	+0.173	**+0.029**
SmolLM2-1.7B	0.782	+0.215	**+0.020**
Qwen2.5-1.5B (256)	0.742	+0.190	**+0.025**
ROCO / Qwen2.5-1.5B	0.693	+0.187	**+0.040**

Table 3. Excess critical-token error against the gold report token, over the full model.

Standard acceptance is lossless but pays full-model compute on every token, whereas a harm-aware variant can accept cheap drafts for ordinary tokens while requiring full-model confirmation only for critical ones, recovering latency without the clinical risk that uniform aggressive acceptance would incur. The risk tagger and stability criterion of TokenTriage are precisely the acceptance policy such a system needs, and our preservation metric is precisely its acceptance rate.

These conclusions should be read with their limits in view, and we state them plainly. Preservation and excess gold-token error are token-level proxies, and the CheXbert experiment measures changes to the finding vector of a counterfactually perturbed gold report, not a radiologist judgment of report correctness; a wrong token that does not move a CheXbert label may still matter clinically, and the converse is possible as well. The logit lens we use to read intermediate layers is a known-noisy probe, and it relies on a final norm exposed by the architecture; we mitigate this only in the sense that every policy is scored against the same probe, so the comparison between policies is fair even where the absolute lens is imperfect. The evidence is confined to chest radiology and to English, the risk tagger is a fixed lexicon rather than a learned or entity-grounded one (though the interface is pluggable), and one model, Phi-3.5-mini, was excluded for a remote-code cache incompatibility, leaving two model families rather than three. Our uncertainty is quantified by report-level bootstraps: the intervals for the main effect do not overlap, but the smaller single-run comparisons are best read as consistency checks rather than independent confirmations. Finally, and most importantly, the negative decoding result in Section 6.5 bounds the scope of the claim: TokenTriage is a tool for the preservation and verification regime, not a drop-in speedup for untrained free generation, and we have tried throughout to report where it does not help as carefully as where it does.

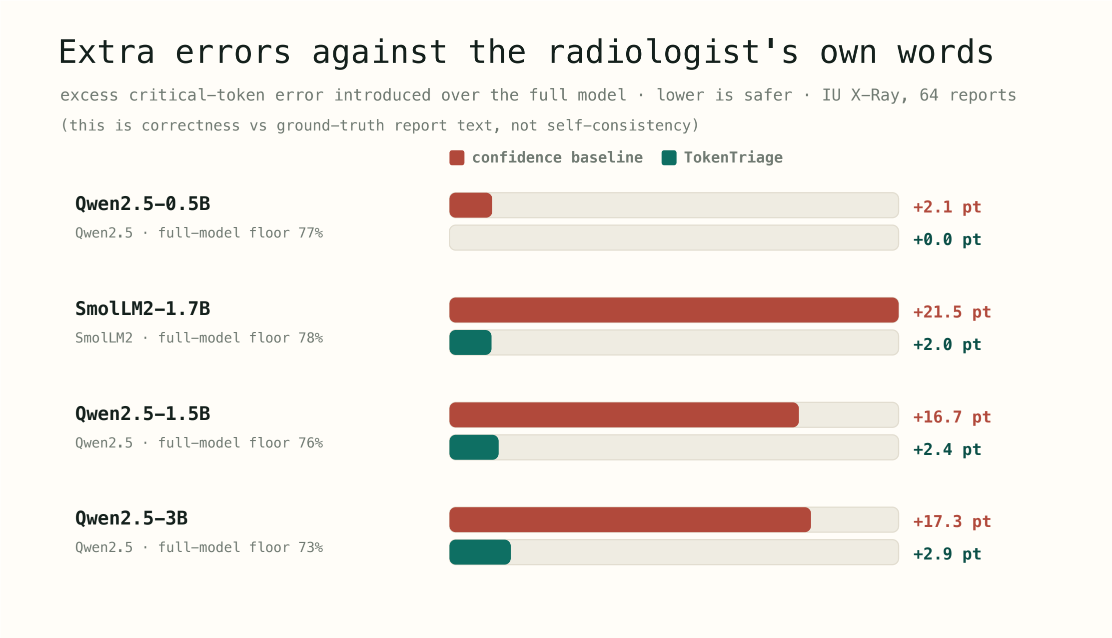

Figure 4. Excess critical-token error against the words of the radiologist. Confidence-only exit (clay) adds 16 to 22 points, while TokenTriage (teal) adds 0 to 3.

	Reports w/ changed finding		Findings flipped	
Run	Entropy	TokenTriage	Entropy	TokenTriage
IU X-Ray (256)	0.699	**0.461**	400	**224**
IU X-Ray (64)	0.750	**0.516**	99	**67**

Table 4. Entity-level correctness (CheXbert, Qwen2.5-1.5B). TokenTriage roughly halves clinical-finding changes relative to confidence-only exit.

8 Conclusion

Efficient clinical inference should ask not only whether a model is confident, but what kind of token is being approximated. Across four open models, two datasets, the words written by the radiologist, and CheXbert findings, confidence-only adaptive inference concentrates its error on clinically critical tokens, and TokenTriage removes most of that error with a training-free routing rule, in the preservation and verification regime where the method applies. The natural next step is to use the same risk signal as the acceptance policy of a harm-aware self-speculative decoder, and to replace the lexicon tagger with a RadGraph- or CheXbert-derived one.

Reproducibility. All traces, policy sweeps, the cross-model, ground-truth, entity-level, and decoding analyses, and the project page are released at https://github.com/DIGlabUAB/token-triage, and every table regenerates from cached traces.

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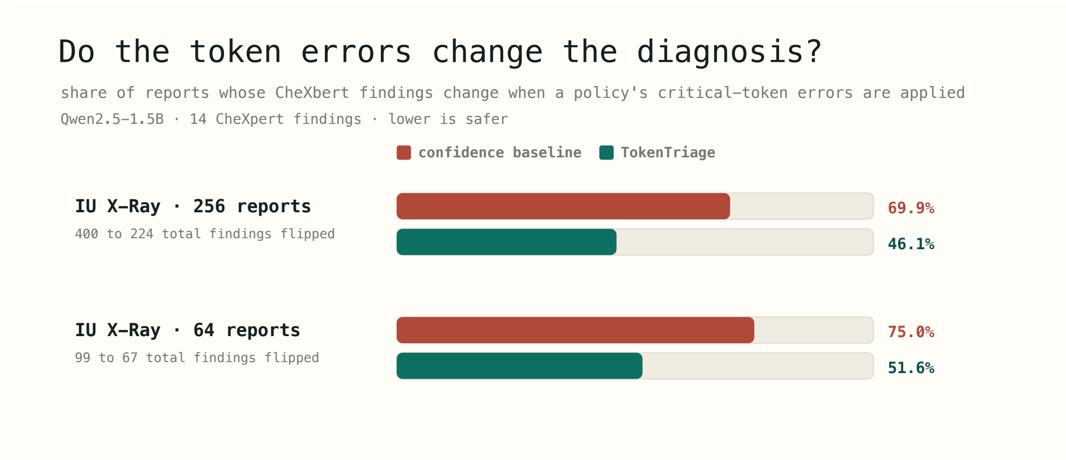

Figure 5. Share of reports whose CheXbert findings change when the critical-token errors of a policy are applied to the gold report.

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